Abstract

Climate change has shifted the timing of major life cycle events across most systems. These shifts may in turn alter how species assemble temporally. While theory predicts that plant species should partition time to reduce resource competition—for example by leafing out at different times—few studies have examined this given the complexity of environmental cues that make timings highly variable across years and sites. Here we examine species- and community-level responses in leafout across cuttings of 47 woody plant species from four sites in North America. By manipulating known temperature and photoperiod cues in controlled environments, then applying a phylogenetic Bayesian model, we estimate the effects of these cues and evolutionary history in shaping leafout across species. We found that each cue caused an advance in the day of leafout, but to varying degrees. Each of these responses, however, were highly similar across sites, exhibiting instead greater variation across species. Differences in species responses to well-studied temperature and photoperiod cues explained only half of the observed variation in leafout date, with the remaining variation explained in part by the estimated phylogenetic structure of leafout, representing species shared evolutionary history. This suggests current models of leafout phenology are missing important variation that may structure species assembly within communities. Identifying these unidentified cues would improve forecasts of phenological shifts and our understanding of their effects on community dynamics.

Introduction

Climate change has led to advances of about three days per decade in the timing of species life history events—phenology (Cohen et al., 2018; Parmesan, 2007; Thackeray et al., 2016). Different types of events, however, are highly variable, with certain events, such as leafout, spanning a period of weeks. Understanding and explaining this variability has become increasingly important as phenology shifts in response to rising temperatures at different rates (Fitter and Fitter, 2002; Fridley, 2012; Menzel et al., 2006; Parmesan, 2007; Yu et al., 2010). Some degree of variability is likely due to differences in climate change itself across space—as some areas warm faster than others and generate larger phenological shifts (Hoegh-Guldberg et al., 2018). Recent research suggests that species-level differences in phenology also drive some of this variation (Vitasse et al., 2009, 2018; Wolkovich and Cleland, 2014; Zohner and Renner, 2014). How these differences compare to other potential sources of variation, and what predicts them, is still largely unknown, but has major implications for predicting how changes in phenology will affect community dynamics and ecosystem services.

Differences in species phenologies can impact community structure and dynamics. Community ecology has long posited that species assemble in part through partitioning time, leading to unique temporal niches that allow species to coexist (Chesson et al., 2004; Grime, 1977). For example, earlier germination or leafout may provide understory species early-access to soil and light resources, before larger canopy species leafout (Heberling et al., 2019; Lee and Ibáñez, 2021). These different temporal niches may also trade-off different pressures, with research suggesting that in seasonal systems species that are active earlier experience greater abiotic pressures—such as spring frost—but avoid the greater competitive pressures for resources that occur later in the season when most species are fully active (Augspurger, 2009; Gotelli and Graves, 1996; Pau et al., 2011; Sakai and Larcher, 1987).

As climate change variably affects the timing of growth, it may alter the temporal order of species in a community and result in novel interactions between species (Cleland et al., 2012; Rudolf, 2019; Tiusanen et al., 2020), thus changing the competitive landscape and synchrony of interactions within communities. How consistent these re-orderings are would depend then on the climatic changes in any one location, the species composition of the community and potentially any local adaptation of phenological cues that may drive population-level differences within species. The relative importance of local adaptation of phenological cues is currently debated, with some research suggesting how plastic versus locally adapted phenological cues are dependent on the specific species, ecosystems, and phenophase (Inouye et al., 2010; Ramirez-Parada et al., 2024).

While recent changes in climate are shaping differences in phenology, for many species their temporal niches may also reflect their evolutionary history. Species traits in communities today are the result of selective pressures acting over evolutionary timescales that include the historical climates and related ancestral phenotypes that were favourable under previous conditions (Wiens et al., 2010). This could produce greater similarity in phenological traits of more closely related species (Davies et al., 2013; Morales-Castilla et al., 2024), which would impact how species respond to future climates.

Spring budburst offers an excellent system to test for species- and community-level patterns in phenology and environmental cues. Budburst of temperate woody plants is well studied and thus known to respond to environmental cues, especially temperatures in the winter and spring—referred to as chilling and forcing, respectively—and daylength or photoperiod (Basler and Körner, 2014; Chuine et al., 2010; Cooke et al., 2012; Laube et al., 2014; Polgar and Primack, 2011). Budburst responses to these cues are generally similar across sites and populations (Aitken and Bemmels, 2016; Zeng and Wolkovich, 2024). Species-level differences, however, are generally much larger than observations of site or population-level variation (but see Deans, 1996; Sogaard et al., 2008, for examples of population-level differences). Decades of phenological research suggests these unique phenologies are likely due to different responses to temperature and photoperiod cues, though they may be also structured by species shared evolutionary history (Davies et al., 2013; Gougherty and Gougherty, 2018; Lechowicz, 1984) and their current environments. Forest understory species, for example, should vary their leafout and budburst early to both gain access to light and soil nutrients prior to canopy closure (Mahall and Bormann, 1978; Muller, 1978), and to reduce competition between species within the understory versus canopy.

Here we test how environmental—temperature and photoperiod—cues and evolutionary relationships shape species-level variation in phenology of forest communities across North America. We leveraged two large-scale growth chamber cutting experiments, in which we vary the temperature cues of chilling (cool temperatures in the fall and winter) and forcing (warmer temperatures, usually in the spring) with photoperiod (Figure 1), with a Bayesian phylogenetic approach to estimate budburst responses across 47 woody species. We collected samples (cuttings) from adult plants at four sites, in eastern and western North America (Figure 1), with pairs of sites on each coast spanning 4 and 6° latitude, respectively. Our dataset includes the diverse assemblages of tree and shrub species found in eastern and western deciduous forests, allowing us to explore differences across both sites and functional groups—including understory shrubs versus canopy trees—in eastern versus western forests.

Materials and Methods

Field sampling

We combined data from two growth chamber studies using dormant branch cuttings of North American deciduous woody plants. Such cutting experiments are a common and powerful approach to infer the responses of adult trees to environmental conditions (Primack et al., 2007; Vitasse et al., 2014). In our first study, we collected samples from two eastern sites—Harvard Forest, Massachusetts, USA (42.55°N, 72.20°W) and St. Hippolyte, Quebec, Canada (45.98°N, 74.01°W), from 20-28 January, 2015 (previously reported in Flynn and Wolkovich, 2018). The second growth chamber study spanned two western sites—Smithers (54.78°N, 127.17°W) and E.C. Manning Park (49.06°N, 120.78°W), British Columbia, Canada, with sampling from 19-28 October 2019 (not previously published). Combining the datasets from these two experiments allowed us to test for spatial variation in the phenology of woody plant communities at a larger geographic scale and make stronger inferences for how different species vary in their cues.

We selected the dominant deciduous species in each site and maximized the number of species occurring across sites. We observed budburst for 47 species spanning our eastern and western forests, with 28 species at our eastern transect and 21 species at our western transect, of which 3 species occurred at both regions (Figure 1, Table S1). Of the species in our eastern transect, there were 13 shrubs and 15 trees, while our western transect included 18 shrub and 4 tree species. At all sites, we tagged 15-20 healthy, individuals for each species in the summer prior to sampling. To reduce ontogenetic effects (Vitasse et al., 2014), we only sampled mature individuals. For trees we sampled individuals greater than 2 m in height and with DBH greater than 4 cm (Natural Resources Canada, 2020), and only sampled shrubs with woody stems. We collected a single cutting from small shrub species, but depending on the size of the sampled tree, we took between 6-16 cuttings, thus collecting samples from 6-20 tagged plants per species. All cuttings were taken from the same relative canopy position, sampling terminal branches from shrubs and the lower canopy from trees using a pole pruner from the ground. We kept samples cold and immediately placed them in erlenmeyer flasks of water upon returning from the field. We conducted the eastern study at the Arnold Arboretum, Boston, MA, USA, and the western study at the University of British Columbia, BC, Canada. We used daily weather data from local weather stations to calculate the field chilling prior to collection (see Table S2).

Growth chamber study

We used growth chambers with two levels of chilling—with either no additional chilling or 30 days at 4°C for our eastern study, and 21 or 56 days of chilling at 4°C for our western study (all dark)—after which we moved plants to one of two levels of forcing—with day and night temperatures that varied to create a cool regime of 15:5°C and a warm regime of 20:10°C—crossed with two photoperiods of either 8 or 12 hours—for a total of 8 distinct treatments in each study.

Our design was similar for both eastern and western sites (Figure 1). We had seven to ten replicates per treatment in our eastern study and eight replicates in our western study, with the exception of *Sorbus scopulina*, for which we only had six replicates per treatment. Our eastern and western studies differed only in the timing of sample collection and thermoperiodicity (the interval of alternating day and night temperatures) in forcing treatments. By collecting samples in our eastern study in late January, they experienced considerable field chilling. We collected the western samples in late October, with most chilling occurring under controlled conditions. In our eastern study, we also set the duration of forcing temperatures to correspond with the photoperiod treatments. While this is a widespread and common approach in phenology chamber studies, it leads to a confounding effect in the thermoperiodicity (Buonaiuto et al., 2023). Thus we applied forcing temperatures for 12 hours independent of the duration of the photoperiod treatments in our western study, and controlled for this difference in our statistical approach (see below). Each experiment was conducted across 8 growth chambers. We additionally rotated cuttings within chambers approximately every two weeks to avoid any chamber effects and changed the water and re-cut each cutting to prevent callusing and keep cuttings healthy. We controlled for small differences in the number of replicates across species in our statistical approach using partial pooling (see section on statistical analysis).

We assessed phenological stages of budburst using the BBCH scale (Finn et al., 2007), adapted for our specific species. We also created photographic guides to help define each stage of the BBCH scale for our specific species (Loughnan and Wolkovich, 2024; Savas et al., 2017). Every 1-3 days, we observed each sample and recorded all phenological stages up to full leaf expansion (defined as code 17 by Finn et al., 2007). Here, we modelled budburst timing, defined as the day of budbreak or shoot elongation (denoted as code 07 by Finn et al., 2007) since forcing treatments began. Our eastern study spanned 82 days, with over 19320 phenological observations, while the western study spanned 113 days and 47844 phenological observations.

Statistical Analysis

We tested for differences in budburst cues across species and sites using a phylogenetic mixed effects model with partial pooling ('shrinkage') across species. This approach accounts for both the evolutionary relatedness of our species and estimates the species-level cues and differences across sites. We obtained species phylogenetic relatedness by pruning the Smith and Brown (2018) megatree of angiosperms (Figure 2).

In our model, we included each of our three cues as continuous variables. To account for the later collection dates and greater amounts of field chilling experienced in our eastern study, we converted our chilling treatments to chill portions for each individual site using local weather station data (Table S2) and the chillR package (v. 0.73.1, Luedeling, 2020). Chill portions are a commonly used chill metric that is calculated using hourly climate data and a Dynamic Model in which cool temperatures lead to the accumulation of chilling, while more accurately estimating chilling accumulation under fluctuating temperatures (Luedeling, 2012, 2020). We calculated chill portions for both the chilling experienced in the field prior to sampling and during the experiment in our 4°C chilling chambers. We also converted forcing temperatures to mean daily temperatures in each treatment to account for differences in thermoperiodicity between the two studies (Buonaiuto et al., 2023). Finally, we z-scored each cue and site using two standard deviations to allow direct comparisons between results across cues (Gelman, 2008).

Our modelling approach allowed us to combine observations of budburst (y_i) across species (sp) to estimate both the effects of species-specific differences in their responses (the rate of change in budburst day per unit of standardized cue, as estimated by the slope, β) to cues (chilling, forcing and photoperiod). By z-scoring the model coefficients, the estimates of the species-level intercepts (α_{sp_i}) represent species differences relative to the mean day of budburst (Gelman et al., 2020). Our full model also estimated effects of sites and all possible two-way interactions between cues, and between cues and sites, to estimate the day of budburst ($\hat{y}$) relative to the first day of forcing conditions and relative to baseline conditions and our most northern site (Smithers).

$$\begin{aligned}
&\hat{y}_i = \alpha_{sp_i} + \beta_{Manning\ park} + \beta_{Harvard\ forest} + \beta_{St.\ Hippolyte} + \\
&\beta_{chilling_{sp_i}} + \beta_{forcing_{sp_i}} + \beta_{photoperiod_{sp_i}} + \\
&\beta_{forcing \times chilling_{sp_i}} + \beta_{forcing \times photoperiod_{sp_i}} + \beta_{chilling \times photoperiod_{sp_i}} + \\
&\beta_{forcing \times Manning\ park_{sp_i}} + \beta_{forcing \times Harvard\ forest_{sp_i}} + \beta_{forcing \times St.\ Hippolyte_{sp_i}} + \\
&\beta_{chilling \times Manning\ park_{sp_i}} + \beta_{chilling \times Harvard\ forest_{sp_i}} + \beta_{chilling \times St.\ Hippolyte_{sp_i}} + \\
&\beta_{photoperiod \times Manning\ park_{sp_i}} + \beta_{photoperiod \times Harvard\ forest_{sp_i}} + \beta_{photoperiod \times St.\ Hippolyte_{sp_i}}
\end{aligned}$$

$$y_i \sim \text{normal}(\hat{y}_i, \sigma^2)$$

The slopes were modelled with partial pooling at the species-level:

$$\begin{aligned}
&\beta_{forcing_{sp}} \sim \text{normal}(\mu_{forcing_{sp}}, \sigma^2_{forcing}) \\
&\ldots \\
&\beta_{photoperiod \times St.Hippolyte_{sp}} \sim \text{normal}(\mu_{photoperiod \times St.Hippolyte_{sp}}, \sigma^2_{photoperiod \times St.Hippolyte_{sp}})
\end{aligned}$$

We included the phylogenetic effect on species-specific differences in budburst timing (α_{sp}) as a variance covariance matrix ($\boldsymbol{V}$) in the parameterization of the normal random vector:

$$\boldsymbol{\alpha_{sp}} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \ldots \\ \alpha_n \end{bmatrix} \text{ such that } \boldsymbol{\alpha} \sim \text{multi-normal}(\mu_\alpha, \boldsymbol{V})$$

The timing of budburst ancestrally is represented as the root value of the means of the multivariate normal distribution. The phylogenetic structuring of budburst is estimated using Pagel's lambda, a phylogenetic scaling parameter where values close to one indicate a phylogenetic signal consistent with Brownian motion and values close to zero the absence of a phylogenetic signal for a given trait (Pagel, 1999; Pearse et al., 2025). For more detail on this method of phylogenetic modeling, see Morales-Castilla et al. (2024). In addition to our main model, we performed a subsequent analysis to check our phylogenetic results were robust to whether we included the phylogenetic effect on the intercept (model shown above) or on the residuals, which is more similar to the common approach of PGLMM

(Pagel, 1999; Pearse et al., 2025); for this we used the same model framework but moved the phylogenetic effect from the intercept to σ_y^2.

We used non-informative priors for each of our model parameters (increasing priors three-fold did not demonstrably change our estimates), and validated the model code using test data prior to analysis. Our model produced $\hat{R}$ values close to 1, well mixed chains, and n_{eff} values that exceeded 10% of the model iterations. Our model was fit using the Stan language with the rstan package (v. 2.26.23, Stan Development Team, 2018) in R (v. 4.3.1, R Development Core Team, 2017).

We present model estimates, relative to baseline or low treatment conditions. This means that we show the average daily temperature for our cooler forcing conditions (which averaged to 8.33°C for our eastern sites and 10°C for our western sites, after accounting for the differences in thermoperiodicity), with a short photoperiod (8 hours photoperiod), and the mean of the lowest chill portions from each population (which is equivalent to approximately 10 weeks of chilling at 4°C) and—when relevant—our most northern site (Smithers). We present results as means and 90% posterior uncertainty intervals from our Bayesian phylogenetic models.

Results

On average, species budburst occurred 30.4 days (uncertainty interval: 26.0, 35.2) after the start of forcing and photoperiod treatments (all estimates are given as mean ± 90% uncertainty intervals, henceforth 'UI,' and budburst dates given as relative to baseline conditions, see methods for more details). Between the earliest and latest budbursting species, the difference in the mean day of budburst was 38.5 and 30.7 days for the eastern and western sites respectively. Budburst of individual species showed distinct differences in their timing and the relative importance of cues, but all cues led to an advance in budburst date. Of the three cues, chilling had the largest effect, with increased chilling advancing budburst (we represent advances with negative numbers, while delays are positive; -14.8, UI: -16.7, -12.9 days per standardized chill portions) and photoperiod the smallest (-3.4, UI: -4.0, -2.7 days per standardized photoperiod, Figure 3a, S1). But we found a large, subadditive interaction between forcing and chilling (8.4, UI: 7.0, 9.8, Figure 3b); meaning that low chilling is offset by high forcing conditions, and vice versa (Table S3).

Overall we found species budburst was moderately phylogenetically structured (Pagel's λ of 0.4, UI: 0.1, 0.8, Fig 2, estimated on the intercept; results were similar when estimated on the residuals; Table

S4). This phylogenetic structure suggests there are latent processes that shape species-level differences in budburst timing beyond the species-specific differences in their responses to temperature (chilling and forcing) and photoperiod. The contribution of these species-specific cue differences to budburst order and timing was small compared to the overall species differences unexplained by cues (represented in the modeling framework as the intercept), with the exact magnitude for which chilling, forcing and photoperiod determine budburst date depending on the level of each cue (e.g., forcing has a much larger effect on budburst timing given a constant daytime temperature of 20°C compared to a constant daytime temperature of 15°C). Across the levels of chilling, forcing and photoperiod we studied, species-specific differences in responses to cues explained between 37.6% and 67.8% of variation for eastern species and between 47.4 and 61.7% for western species, with the phylogenetically structured intercept explaining the rest (Figure 2).

We also found the rank order of species' budburst timings changed more when estimated using higher chilling (which averaged across all sites, is equivalent to approximately 15 weeks chilling at 4°C) and forcing, and longer photoperiods than under lower chilling and forcing, and shorter photoperiods (Figure S2). The budburst ranking for the three species that occurred in both transects remained relatively consistent across our eastern and western sites (Figure 4), with only *Alnus incana* in the western sites experiencing a large change in rank with cues (Figure S2). In comparing the earliest and latest budbursting species, we found relatively small differences in cues (Figure 4). For example, an early budbursting shrub, *Lyonia ligustrina*, had chilling and forcing cue estimates of -14.7 and -11.5 respectively, which are comparable to the cue estimates of -15.7 and -11.5 for the much later tree, *Quercus alba*. Yet the model predicts these species to have a 15.1 day difference in day of budburst, highlighting how the timing of budburst across species was also strongly related to species-level differences (intercept values) independent of their responses to chilling, forcing and photoperiod (see Figure 5).

Despite our expectations that species responses should differ across the canopy layer, with shrubs being less sensitive to cues than trees, we did not find substantial differences across these two functional groups (Figure S3). Shrubs, like *Cornus stolonifera*, budburst early and showed small responses to chilling and forcing (Figure 4b, d). But 36.7% of the shrubs, including *Menziesia ferruginea* and *Symphoricarpos alba*, exhibited the opposite response and budburst relatively late (Figure 5, S3). Similarly for trees, some species did match our predictions, exhibiting stronger cues, but 23.5% of species budburst earlier than expected. While specific tree species, such as *Quercus velutina*, did have larger chilling and photoperiod responses as predicted (Figure 4a, e), and *Fagus grandifolia* had the largest photoperiod response (Figure 4e), but overall we did not find clear differences between the cues of trees and shrubs across the four sites.

Finally, we observed negligible site-level effects compared to differences between cues (Figure 3, Table S3). Treatments that received a month less of chilling (low chilling), that were 5°C cooler (low forcing) with shorter photoperiods, all budburst later compared to the more chilled, warmer and long photoperiod treatments across all sites and across transects (Figure S1 a-c, shown for baseline conditions, see Table S3 for model output). Overall budburst dates did not differ between sites, though eastern sites budburst marginally earlier (day 35.8, UI: 28.3, 42.3) compared to the western sites (day 48.1, UI: 40.7, 54.5, Figure 5, see also Table S3 for model output). This could be due to the earlier collection date of our western species, which reduced the field chilling they received relative to our eastern samples (though see corrections for this in the statistical analysis section of our methods).The substantial overlap in budburst dates overall, and when estimated with low and high cues (see Figure S1) supports that the forest communities at these different sites are highly similar, despite the large distances and differences in species compositions between them.

Discussion

To accurately forecast shifts in phenology and its impact on ecological communities requires understanding which environmental cues are most important at the population-, species-, functional-group and community-levels. Our results support decades of research findings that budburst timing advances in response to the well-studied major temperature cues of chilling and forcing (with the greatest responses to chilling and forcing respectively, see also Ettinger et al., 2020; Flynn and Wolkovich, 2018) and longer photoperiods. But using our modelling approach allowed us to estimate the extent to which these species-specific responses to these cues do—or do not—lead to differences in species timings that may structure temporal niches within communities. Perhaps surprisingly, we found a large portion (one-third to one-half) of variation in species budburst timings was unexplained by different responses to chilling, forcing and photoperiod. Our results show that some of this variation is due to species evolutionary history, but not all, suggesting fundamental gaps in our model of one of the best studied and most important phenological events—woody plant budburst.

Differences in species responses to temperature and photoperiod were similar across sites and functional groups, suggesting that understanding the drivers of variation across species may be an important task to improve forecasting. Our study included 47 of the dominant woody species from deciduous forests across North America from both shrub and tree functional groups, with a mix of early and late budbursting phenologies. Despite the high degree of diversity and spatial scale represented in our study, we found no detectable differences in the responses to temperature and photoperiod at our different sites, even though sites span different coasts and over 6° of latitude. This contrasts with research that suggests latitudinal trends in photoperiod and temperature may impose unique selective pressures, and thus may drive site-level differences in responses to cues (Keller et al., 2011). Differences in spring phenology have been found across latitudinal gradients by studies using *in situ* phenological data, with some studies finding greater rates of phenological change at higher latitudes across diverse taxa (Post et al., 2018), and other studies, focused on specific taxa such as trees, finding evidence for greater phenological responses at lower latitudes (Alecrim et al., 2023). In contrast, studies using common gardens or conducted across elevation gradients also find phenological differences between populations, but attribute them largely to differences in local microclimates (McDonough MacKenzie et al., 2018; Stinston, 2005; Vitasse et al., 2013) or adaptation of species to shorter growing seasons (Gugger et al., 2015), and with genetic variation having a weaker effect (Vitasse et al., 2009, 2013). Many of these studies, however, are based on observations in the field, with confounding differences in study duration and start dates, variable methodologies, and geographic extent that make fully attributing variation to local adaptation difficult. The lack of site-level effects that we observed—using experiments to estimate responses—suggests budburst cues are not shaped by local environmental conditions, at least at our study scale.

Community composition and interspecific variation in phenology

The variation across species in their responses to environmental cues has the potential to create large differences in species temporal niches and ecological roles. Species ranged from early to late budburst timing in our experimental conditions, spanning a similar period as observed in natural communities (Lechowicz, 1984; Richardson, A.D., O'Keefe, 2009). This suggests our experiment captured a realistic breadth in temperate forest budburst. Furthermore, all our focal species responded to each environmental cue, with large responses to chilling and comparatively small responses to photoperiod—trends consistent with previous studies (Ettinger et al., 2020; Flynn and Wolkovich, 2018). Cues, however, had complex interactions that appear potentially advantageous under warming climates (Caffarra and Donnelly, 2011; Flynn and Wolkovich, 2018; Heide, 1993). The interaction between chilling and forcing may ensure that species budburst even if warmer winters cause insufficient chilling, but will require additional forcing. But despite these differences in species responses to cues, we did not find clear, generalizable trends across functional groups.

Shrubs and trees differ greatly in their physiology, filling different ecological niches. Most of our trees budburst later than our study's shrubs, a relative order also found by previous studies (Gill et al., 1998; Panchen et al., 2014; Polgar et al., 2014). But about a quarter of our trees also budburst early, with timing more similar to that of shrubs. Similarly, we found a third of shrubs budburst at similar times as the majority of our trees. These deviations from our expectations for how shrubs and trees partition their budburst timing suggests there is more nuance to these patterns. As tree species advance phenologically—resulting in earlier canopy closure and reduced light availability (Donnelly and Yu, 2019; Heberling et al., 2019)—this subset of shrubs with small cues may have reduced fitness. Our findings suggest, however, that many shrubs will also advance phenologically and have the potential to maintain their relative temporal niche.

Community assembly in responses to cues versus evolutionary history

Our study included the three cues—chilling, forcing and photoperiod—commonly thought to determine budburst timing, but over a third of the total variation was not explained by these cues (Figure 4), suggesting the general model of budburst may be incomplete. This, paired with the observed phylogenetic structure in budburst timing, hints at unidentified latent traits still missing from our understanding of the processes shaping spring phenology (Davies et al., 2019; Webb et al., 2002).

Latent traits could reflect other cue or trait relationships indicative of species growth strategies or key traits that may facilitate greater resource use, or the greater competitive abilities needed when budbursting later (Grime, 1977; Inouye et al., 2010). For example, species that budburst later in the season may have more conservative growth strategies, possessing traits that allow them to compete for limited soil resources and light, such as low specific leaf areas or higher wood densities (Chave et al., 2009; Wright et al., 2004). But our results suggest that evolutionary history and species ancestral phenotype also have some effect on species timings and cues. Species traits are shaped over 'deep' timescales, with previous evolution also influencing responses to the present climate. Phenological trends could thus be an indication of limited evolutionary time to adapt (Lechowicz, 1984), or shaped by selection on other important traits that are not phylogenetically structured. Incorporating both phenology and evolutionary history into a broader trait framework could thus provide insights into traits that correlate with budburst timing, and—ultimately—the current and evolutionary drivers that select for species phenotypes and possibly shape species adaptive potential to future climates.

Predicting budburst under future climates

As climate change leads to greater temperature cues, we are already observing advances in species budburst within diverse ecological communities. These changes are likely to have cascading impacts on species interactions, with greater overlap in growth timing potentially increasing competition for pollinators, resources, or light. Over time, such impacts could alter species performance and species assemblages. The greatest ecological impacts are likely to occur in high latitude communities that are warming the most and the fastest (Hoegh-Guldberg et al., 2018). Our results on the effects of cues on rank order suggests this extreme warming could alter the order with which species budburst, further creating novel species interactions relative to communities experiencing smaller changes in cues. Yet our ability to predict these changes is often limited by a lack of available data for most species and biases in which species have been studied to date.

In addition to our findings providing insights into the phenological cues of understudied temperate forests communities, our approach encompasses several recent advances in forecasting phenological responses to climate change. By accounting for shared evolutionary history when estimating species-level variability, our model better estimates differences in cue responses in comparison to traditional methods that treat all species as exchangeable and produce estimates for species from poorly sampled clades that are influenced by well-sampled ones (Morales-Castilla et al., 2024). This allows us to make better and more meaningful forecasts of future phenological shifts for species that are understudied, rare, or challenging to study. Secondly, the consistency of cue responses we observed across communities indicates that species responses to future climates may be predictable at regional spatial scales. This suggests our understanding of forest community responses to climate change may benefit more from studies that encompass high taxonomic diversity—encompassing a broader extent of species-level variation—as opposed to studies that focus on a few species over larger spatial scales.

While our findings have applications to plant communities, our analytical approach is widely applicable to forecasting diverse species responses to climate change. The phylogenetic model we used here could be easily applied to other phenological events or suites of species from across the tree of life. By combining species-rich and community-wide approaches with phylogenetic relationships we can better understand the impacts of evolutionary history and current ecological processes in shaping species phenology under a changing climate.

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Author Contribution Statement

Both D.L. and E.M.W. conceived the study, contributed to the analysis and code and contributed to the writing and revision of the manuscript. DL led field and experimental data collection for western sites.

Competing Interests Statement

The authors have no competing interests.

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Captions

Figure 1: We combined data from two controlled environment studies of temperature and photoperiod cues for budburst in temperate woody species **a,** from two forest communities (sites) in eastern North America that span a 4° latitudinal gradient and **b,** two communities (sites) in western North America that span a 6° latitudinal gradient. Combined, our dataset includes 28 tree and shrub species from eastern sites and 21 species from western sites, with three species occurring across all four sites. Both studies used a full-factorial design with two levels each of chilling treatments (consisting of a combination of field and different durations of chamber chilling), forcing and photoperiod, for a total of eight unique treatments. Observations of budburst were made during the forcing treatment, using the BBCH scale to classify stages of bud development (see Loughnan and Wolkovich, 2024; Savas et al., 2017, for guides we made for these species). This figure was created by Deirdre Loughnan using adapted icons of a barren tree by Gordon Dylan Johnson (2015) and thermometer by pnx (2014) from Openclipart and icons of a beaker by Brett Patrick Casey (2018) and snowflake by Mrmw (2008) from Wikimedia Commons. All of which are in the public domain.

Figure 2: Tree (denoted with an asterisk) and shrub species differences were accounted for by including phylogenetic effects on the intercept of a model estimating days to budburst after the start of forcing treatments (simplified to 'days' here). This approach estimated a phylogenetic scaling parameter for the intercept of 0.4 (where values close to one indicate a phylogenetic effect consistent with Brownian motion and values close to zero the absence such an effect). Included for each tip of the phylogeny is the species-level intercept value in brackets, along with the names of major the clades represented in the tree.

Figure 3: **a,** Posterior distributions of estimated changes in budburst per unit change in chilling, forcing, and photoperiod with individual site-level effects. Symbols denote the eastern (triangle) and western (square) sites and the darker point color our more northern sites. Points represent the mean, thicker lines the 50% uncertainty interval, and the thin lines the 90% uncertainty interval. **b,** Cues interacted to produce a subadditive effect, where delays in budburst under low chilling was offset by high forcing conditions, and vice versa. To make results comparable across cues we standardized chilling, forcing, and photoperiod via z-scores using two standard deviations, see methods for further details.

Figure 4: Estimated change in species budburst day per unit of standardized chilling (**a,b**), forcing (**c,d**), and photoperiod (**e,f**), ordered by estimated budburst dates for both the eastern (**a,c,e**) and western sites (**b,d,f**) based on our model estimates. Cues are plotted on differing y-axis scales to better

depict species differences across cues. For each species, the points represent the mean, thinner error bars represent the 90% uncertainty interval, and thicker bars represent the 50% uncertainty interval. Points for the three species that occur in both the eastern and western transects are depicted using darker colours. Tree species are labeled above the x-axis and shrub species below the x-axis. To make the results across cues directly comparable, we standardized predictors (here, chilling, forcing, and photoperiod) via z-scores using two standard deviations, see methods for further details. See Table S1 for full species names.

Figure 5: Comparisons of estimated day of budburst for tree (labeled above the x-axis) and shrub species (labeled below the x-axis) based on the full model (intercept plus all cues), shown as white (open) symbols, versus the intercepts only (without effects of chilling, forcing, and photoperiod), shown as filled black symbols. Points are ordered according to differences in species estimated budburst dates for both the eastern (**a**), and western sites (**b**), with low cue conditions, depicted as circles, and high cue conditions, depicted as triangles. See Table S1 for full species names.